Year 8 Science

Elements and Compounds

1. Determine which of the following are true or false. For each statement which is false, substitute for the underlined word, the word which makes the statement correct.

   (a) Matter is anything that takes up space.
   (b) Most elements known today are man-made.
   (c) A carbon granule is the smallest possible part of the element carbon that has all the properties of carbon.
   (d) CA is probably the symbol for the element, calcium.
   (e) In Japan, chemical symbols are the same as in the United States.
   (f) The chemical symbol for the element, argon, is AR.
   (g) The chemical symbol for the element, bromine, could be Bro.
   (h) The second letter of a chemical symbol is always lower case.
   (i) A compound can not be broken down into simpler substances by ordinary means.
   (j) Table salt (sodium chloride) has properties very similar to the elements, sodium and chlorine.
   (k) The formula, NH_3, tells you that there are 3 nitrogen and 3 hydrogen atoms.
   (l) The formula, H_2O, represents one molecule of water.
   (m) The smallest part of the compound, carbon dioxide, that can exist is a molecule of carbon dioxide.

2. Write in the words that best complete each statement.

   (a) An element is the simplest form of _____.
   (b) The ancient Greeks believed there were only 4 elements: _____, _____, _____ and _____.
   (c) The smallest possible part of the element, calcium, is a calcium _____.
   (d) If you heat two different substances and find that substance 1 breads down into two new materials while substance 2 doesn't change, then substance 1 must be a _____ while substance 2 is an _____.

(e) If you heat mercuric oxide, it will break down to form _____ and _____.

(f) A chemical formula tells you which _____ are present in a material.

3. Complete the table below.

Name	Symbol
Sulfur	
	C
Oxygen	
	Cl
Mercury	
Uranium	
	H
	Fe
	Al
Bromine	